

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 2 — Atoms, Molecules & Stoichiometry (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	2.1 Relative Masses · 2.2 Mole Concept · 2.3 Formulas & Hydrates · 2.4 Titrations & Reacting Masses · High-Frequency Core Repeats
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

2.1 Relative Masses of Atoms & Molecules	— Carbon-12 standard, relative isotopic mass, relative atomic mass A_r , relative molecular mass M_r , relative formula mass M_f
2.2 The Mole & Avogadro Constant	— The mole as amount of substance, Avogadro constant $L = 6.02 \times 10^{23} \text{ mol}^{-1}$, mass of single atoms, molar volume
2.3 Formulas (Empirical, Molecular, Hydrated)	— Determining empirical and molecular formulas from percentage compositions, combustion analysis masses (C, H, O, N, S, Cl, Br, I)
2.4 Reacting Masses, Titrations & Atom Economy	— Stoichiometric calculations, limiting reagents, theoretical vs actual yields, percentage atom economy (addition reactions)
High-Frequency Core Repeats	— The 10 most frequently examined Cambridge Paper 1 questions on Stoichiometry (Q101 – Q110).

SUBTOPIC 2.1 — RELATIVE MASSES OF ATOMS AND MOLECULES (Q1 – Q25)

1 [EASY] Standard Reference for Atomic Masses — 9701/12/M/J/23/Q7

[1]

Which isotope is used as the universal standard reference against which all relative atomic, isotopic, and molecular masses are measured?

- [A] Hydrogen-1, assigned a relative mass of exactly 1 unit
- [B] Carbon-12, assigned a relative mass of exactly 12 units
- [C] Oxygen-16, assigned a relative mass of exactly 16 units
- [D] Carbon-14, assigned a relative mass of exactly 14 units

2 [EASY] Relative Molecular Mass of Hydrated Salt — 9701/11/O/N/23/Q8

[1]

What is the relative formula mass of hydrated iron(II) sulfate crystals, $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$?

[Ar: Fe = 55.8, S = 32.1, O = 16.0, H = 1.0]

- [A] 151.9
- [B] 241.9
- [C] 295.9
- [D] 277.9

3 [EASY] Distinguishing Relative Molecular vs Formula Mass — 9701/13/M/J/22/Q3

[1]

For which substance is the term 'relative molecular mass' strictly applicable?

- [A] Carbon dioxide, CO_2
- [B] Sodium chloride, NaCl
- [C] Silicon dioxide, SiO_2
- [D] Magnesium oxide, MgO

4 [HARD] Molecular Masses of Isotopically Substituted Methane — 9701/12/F/M/24/Q2

[1]

Carbon exists as ^{12}C and ^{13}C . Hydrogen exists as ^1H and ^2H (deuterium, D).

How many different values of relative molecular mass are possible for a methane molecule (CH_4)?

- [A] 5
- [B] 8
- [C] 10
- [D] 6

5 [EASY] Relative Formula Mass of Ammonium Dichromate(VI) — 9701/11/M/J/23/Q8

[1]

What is the relative formula mass of ammonium dichromate(VI), $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$?

[Ar: N = 14.0, H = 1.0, Cr = 52.0, O = 16.0]

- [A] 252.0
- [B] 236.0
- [C] 218.0
- [D] 268.0

6 [EASY] Identifying Compound with $M_r = 60$ — 9701/12/O/N/23/Q7

[1]

Which organic compound has a relative molecular mass, M_r , of exactly 60.0?

[Ar: C = 12.0, H = 1.0, O = 16.0]

- [A] Propan-1-ol, $\text{C}_3\text{H}_7\text{OH}$
- [B] Methanoic acid, HCOOH
- [C] Ethanal, CH_3CHO
- [D] Ethanoic acid, CH_3COOH

7 [HARD] Percentage by Mass of Nitrogen in Fertilizer — 9701/13/O/N/22/Q5

[1]

Which fertilizer contains the highest percentage of nitrogen by mass?

[Ar: N = 14.0, H = 1.0, C = 12.0, O = 16.0, S = 32.1, P = 31.0]

- [A] Ammonium nitrate, NH_4NO_3 (Mr = 80.0)
- [B] Urea, $(\text{NH}_2)_2\text{CO}$ (Mr = 60.0)
- [C] Ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$ (Mr = 132.1)
- [D] Potassium nitrate, KNO_3 (Mr = 101.1)

8 [HARD] Relative Molecular Mass of a Gaseous Hydrocarbon — 9701/12/M/J/22/Q6

[1]

A 0.28 g sample of a gaseous alkene occupies 120 cm³ at room temperature and pressure (where 1 mol = 24.0 dm³).

What is the relative molecular mass of the alkene, and which alkene is it?

[Ar: C = 12.0, H = 1.0]

- [A] Mr = 42, propene (C_3H_6)
- [B] Mr = 28, ethene (C_2H_4)
- [C] Mr = 56, but-1-ene (C_4H_8)
- [D] Mr = 70, pent-1-ene (C_5H_{10})

9 [HARD] Formula Mass of Alum Double Salt — 9701/11/F/M/23/Q4

[1]

Potash alum is a double sulfate salt with the formula $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$.

What is its relative formula mass?

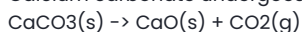
[Ar: K = 39.1, Al = 27.0, S = 32.1, O = 16.0, H = 1.0]

- [A] 258.3
- [B] 474.3
- [C] 456.3
- [D] 492.3

10 [EASY] Mass of Metal Oxide from Thermal Decomposition — 9701/13/O/N/21/Q6

[1]

Calcium carbonate undergoes thermal decomposition according to:



What mass of CaO is produced from the complete decomposition of 50.0 g of pure CaCO_3 ?

[Ar: Ca = 40.1, C = 12.0, O = 16.0]

- [A] 28.0 g
- [B] 56.1 g
- [C] 22.0 g
- [D] 14.0 g

11 [HARD] Mass of Water in Hydrated Copper Sulfate — 9701/12/M/J/24/Q5

[1]

In an experiment, 2.495 g of blue hydrated copper(II) sulfate crystals, $\text{CuSO}_4 \cdot x\text{H}_2\text{O}$, were heated strongly to drive off all water of crystallization. The mass of white anhydrous CuSO_4 remaining was 1.595 g.

What is the value of x ?

[Ar: Cu = 63.5, S = 32.1, O = 16.0, H = 1.0]

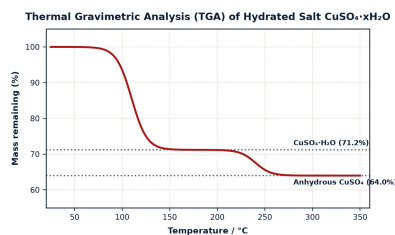


Fig. 11.1: Thermogravimetric thermal mass loss curve of hydrated copper sulfate.

- [A] 4
- [B] 6
- [C] 7
- [D] 5

12 [EASY] Percentage of Oxygen in Magnesium Nitrate — 9701/11/O/N/22/Q6

[1]

What is the percentage by mass of oxygen in anhydrous magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$?

[Ar: Mg = 24.3, N = 14.0, O = 16.0]

- [A] 32.4%
- [B] 54.2%
- [C] 64.7%
- [D] 72.1%

13 [HARD] Relative Formula Mass of Barium Hydroxide Octahydrate — 9701/13/F/M/22/Q3

[1]

What is the relative formula mass of barium hydroxide octahydrate, $\text{Ba}(\text{OH})_2 \cdot 8\text{H}_2\text{O}$?

[Ar: Ba = 137.3, O = 16.0, H = 1.0]

- [A] 315.3
- [B] 171.3
- [C] 285.3
- [D] 331.3

14 [HARD] Identifying Gaseous Oxide from Density at r.t.p. — 9701/12/O/N/21/Q5

[1]

A gaseous oxide of nitrogen has a density of 1.917 g dm^{-3} at room temperature and pressure (molar volume = $24.0 \text{ dm}^3 \text{ mol}^{-1}$).

What is the formula of this oxide?

[Ar: N = 14.0, O = 16.0]

- [A] N_2O (Mr = 44.0)
- [B] NO (Mr = 30.0)
- [C] NO_2 (Mr = 46.0)
- [D] N_2O_4 (Mr = 92.0)

15 [EASY] Relative Formula Mass of Calcium Phosphate — 9701/11/M/J/21/Q5

[1]

What is the relative formula mass of calcium phosphate, $\text{Ca}_3(\text{PO}_4)_2$?

[Ar: Ca = 40.1, P = 31.0, O = 16.0]

- [A] 279.3
- [B] 310.3
- [C] 215.2
- [D] 341.3

16 [HARD] Percentage of Water in Washing Soda Crystals — 9701/13/M/J/23/Q5

[1]

What is the percentage by mass of water of crystallization in washing soda, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$?

[Ar: Na = 23.0, C = 12.0, O = 16.0, H = 1.0]

- [A] 37.1%
- [B] 45.5%
- [C] 55.0%
- [D] 62.9%

17 [EASY] Mass of One Mole of Naturally Occurring Chlorine Gas — 9701/12/M/J/23/Q8

[1]

What is the mass of one mole of naturally occurring chlorine gas, Cl_2 ?

[Ar: Cl = 35.5]

- [A] 35.5 g
- [B] 70.0 g
- [C] 71.0 g
- [D] 72.0 g

18 [HARD] Comparing Percentage by Mass of Carbon in Fuels — 9701/11/O/N/23/Q9

[1]

Which hydrocarbon fuel has the highest percentage of carbon by mass?

[Ar: C = 12.0, H = 1.0]

- [A] Ethyne, C_2H_2
- [B] Ethene, C_2H_4
- [C] Ethane, C_2H_6
- [D] Methane, CH_4

19 [HARD] Molar Mass of a Volatile Liquid via Dumas Method — 9701/13/O/N/23/Q5

[1]

In a syringe experiment, 0.160 g of a volatile liquid vaporised completely to produce 65.0 cm³ of gas at 100 °C (373 K) and 101 kPa.

Using $pV = nRT$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$), what is the relative molecular mass of the liquid?

- [A] 58
- [B] 88
- [C] 102
- [D] 76

20 [HARD] Formula Mass of Mohr's Salt — 9701/12/F/M/24/Q3

[1]

Mohr's salt is ammonium iron(II) sulfate hexahydrate, $(\text{NH}_4)_2\text{Fe}(\text{SO}_4)_2 \cdot 6\text{H}_2\text{O}$.

What is its relative formula mass?

[Ar: N = 14.0, H = 1.0, Fe = 55.8, S = 32.1, O = 16.0]

- [A] 284.0
- [B] 392.0
- [C] 374.0
- [D] 412.0

21 [EASY] Identifying Gaseous Hydrocarbon from Mr — 9701/11/F/M/22/Q3

[1]

A gaseous alkane has a relative molecular mass, M_r , of 44.0.

What is the formula of this alkane?

[Ar: C = 12.0, H = 1.0]

- [A] C_4H_{10} (butane)
- [B] C_3H_8 (propane)
- [C] C_2H_6 (ethane)
- [D] C_5H_{12} (pentane)

22 [HARD] Percentage of Chlorine in Bleaching Powder — 9701/12/M/J/21/Q5

[1]

Bleaching powder can be represented as calcium hypochlorite chloride, $\text{Ca}(\text{OCl})\text{Cl}$.

What is the percentage of chlorine by mass in pure $\text{Ca}(\text{OCl})\text{Cl}$?

[Ar: Ca = 40.1, O = 16.0, Cl = 35.5]

- [A] 28.0%
- [B] 45.2%
- [C] 62.1%
- [D] 55.9%

23 [EASY] Relative Formula Mass of Lead(II) Nitrate — 9701/13/M/J/21/Q4

[1]

What is the relative formula mass of lead(II) nitrate, $\text{Pb}(\text{NO}_3)_2$?

[Ar: Pb = 207.2, N = 14.0, O = 16.0]

- [A] 331.2
- [B] 269.2
- [C] 315.2
- [D] 347.2

24 [EASY] Relative Molecular Mass of Aspirin — 9701/11/O/N/21/Q6

[1]

Aspirin has the molecular formula $\text{C}_9\text{H}_8\text{O}_4$. What is its relative molecular mass?

[Ar: C = 12.0, H = 1.0, O = 16.0]

- [A] 168.0
- [B] 194.0
- [C] 180.0
- [D] 152.0

25 [HARD] Calculating Water of Crystallization for Epsom Salt — 9701/12/M/J/22/Q7

[1]

A sample of hydrated magnesium sulfate, $\text{MgSO}_4 \cdot x\text{H}_2\text{O}$, contains 51.2% water of crystallization by mass.

What is the integer value of x?

[Ar: Mg = 24.3, S = 32.1, O = 16.0, H = 1.0]

- [A] 7
- [B] 5
- [C] 6
- [D] 8

SUBTOPIC 2.2 — THE MOLE AND THE AVOGADRO CONSTANT (Q26 – Q50)

26 [EASY] Definition of the Avogadro Constant — 9701/12/M/J/23/Q9

[1]

Which statement correctly defines the Avogadro constant, L ?

- [A] The mass in grams of one mole of a substance.
- [B] The volume occupied by one mole of any gas at room temperature and pressure.
- [C] The number of protons present in 1.00 g of hydrogen gas.
- [D] The number of specified particles (atoms, molecules, or ions) in one mole of any substance ($6.02 \times 10^{23} \text{ mol}^{-1}$).

27 [HARD] Sample Containing the Greatest Number of Molecules — 9701/11/O/N/23/Q10

[1]

Which sample contains the greatest total number of molecules?

[Ar: H = 1.0, C = 12.0, N = 14.0, O = 16.0]

- [A] 2.0 g of methane gas, CH_4
- [B] 1.0 g of hydrogen gas, H_2
- [C] 4.0 g of oxygen gas, O_2
- [D] 7.0 g of nitrogen gas, N_2

28 [HARD] Sample Containing Greatest Number of Atoms — 9701/13/M/J/22/Q4**[1]**

Which sample contains the greatest total number of individual atoms?

[Ar: H = 1.0, C = 12.0, O = 16.0, Ne = 20.2]

- [A] 18.0 g of water, H₂O
- [B] 44.0 g of carbon dioxide, CO₂
- [C] 16.0 g of methane, CH₄
- [D] 20.2 g of neon, Ne

29 [HARD] Number of Electrons in 1.8 g of Water — 9701/12/M/J/22/Q8**[1]**How many electrons are present in 1.80 g of pure liquid water, H₂O?[Avogadro constant $L = 6.02 \times 10^{23} \text{ mol}^{-1}$; Ar: H = 1.0, O = 16.0]

- [A] 6.02×10^{22}
- [B] 6.02×10^{23}
- [C] 1.08×10^{24}
- [D] 6.02×10^{24}

30 [HARD] Total Number of Ions in Calcium Chloride Solution — 9701/11/M/J/23/Q9**[1]**A solution contains 11.1 g of calcium chloride, CaCl₂, completely dissolved in water.

How many total ions (calcium ions + chloride ions) are present in this solution?

[$L = 6.02 \times 10^{23} \text{ mol}^{-1}$; Ar: Ca = 40.1, Cl = 35.5]

- [A] 1.81×10^{23}
- [B] 6.02×10^{22}
- [C] 1.20×10^{23}
- [D] 2.41×10^{23}

31 [EASY] Volume Occupied by Gaseous Carbon Dioxide — 9701/13/O/N/23/Q6**[1]**

What volume is occupied by 11.0 g of carbon dioxide gas at room temperature and pressure?

[Molar volume at r.t.p. = $24.0 \text{ dm}^3 \text{ mol}^{-1}$; Ar: C = 12.0, O = 16.0]

- [A] 12.0 dm³
- [B] 24.0 dm³
- [C] 3.0 dm³
- [D] 6.0 dm³

32 [HARD] Mass of a Single Atom of Lead — 9701/12/O/N/22/Q7**[1]**

What is the approximate mass of a single atom of lead (Ar = 207.2)?

[$L = 6.02 \times 10^{23} \text{ mol}^{-1}$]

- [A] $2.07 \times 10^{-21} \text{ g}$
- [B] $1.66 \times 10^{-24} \text{ g}$
- [C] $3.44 \times 10^{-22} \text{ g}$
- [D] $6.02 \times 10^{-23} \text{ g}$

33 [HARD] Number of Covalent Bonds in 16 g of Methane — 9701/11/F/M/24/Q4**[1]**How many covalent C-H bonds are present in 16.0 g of methane, CH₄?[$L = 6.02 \times 10^{23} \text{ mol}^{-1}$; Ar: C = 12.0, H = 1.0]

- [A] 2.41×10^{24}
- [B] 6.02×10^{23}
- [C] 1.20×10^{24}
- [D] 3.01×10^{24}

34 [EASY] Number of Chloride Ions in Solid Barium Chloride — 9701/12/M/J/21/Q6

[1]

How many chloride ions are present in 0.050 moles of solid barium chloride, BaCl₂?

[L = 6.02 × 10²³ mol⁻¹]

- [A] 3.01 × 10²²
- [B] 1.20 × 10²³
- [C] 6.02 × 10²²
- [D] 1.81 × 10²³

35 [EASY] Density of Gas at r.t.p. — 9701/13/M/J/24/Q5

[1]

What is the density in g dm⁻³ of sulfur dioxide gas, SO₂, at room temperature and pressure?

[Molar volume at r.t.p. = 24.0 dm³ mol⁻¹; Ar: S = 32.1, O = 16.0]

- [A] 1.33 g dm⁻³
- [B] 2.67 g dm⁻³
- [C] 3.20 g dm⁻³
- [D] 0.375 g dm⁻³

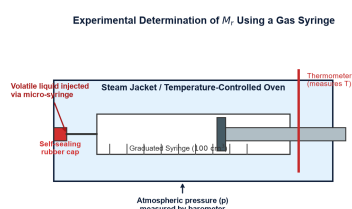
36 [HARD] Number of Molecules in Gas Syringe Reading — 9701/11/O/N/22/Q7

[1]

A gas syringe collects 48.0 cm³ of carbon monoxide gas, CO, at room temperature and pressure.

How many molecules of carbon monoxide are present in the syringe?

[L = 6.02 × 10²³ mol⁻¹; molar volume at r.t.p. = 24000 cm³ mol⁻¹]



- [A] 6.02 × 10²⁰
- [B] 2.41 × 10²¹
- [C] 1.20 × 10²²
- [D] 1.20 × 10²¹

37 [HARD] Comparing Quantities Containing 1.20 × 10²⁴ Atoms — 9701/12/F/M/23/Q5

[1]

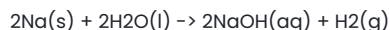
Which quantity contains exactly 1.204 × 10²⁴ total atoms?

[L = 6.02 × 10²³ mol⁻¹]

- [A] 2.0 moles of ozone gas, O₃
- [B] 0.5 moles of methane gas, CH₄
- [C] 1.0 mole of chlorine gas, Cl₂
- [D] 1.0 mole of helium gas, He

38 [EASY] Moles of Hydrogen Gas from Reaction of Sodium with Water — 9701/13/O/N/21/Q5

[1]



How many moles of hydrogen gas are liberated when 0.46 g of sodium reacts completely with excess water?

[Ar: Na = 23.0]

- [A] 0.010 mol
- [B] 0.020 mol
- [C] 0.005 mol
- [D] 0.040 mol

39 [HARD] Number of Protons in 1.0 g of Deuterium Gas — 9701/11/M/J/22/Q7

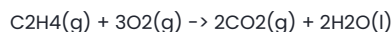
[1]

How many protons are present in 1.00 g of deuterium gas, 2H_2 ?[$L = 6.02 \times 10^{23} \text{ mol}^{-1}$]

- [A] 6.02×10^{23}
[B] 1.20×10^{24}
[C] 1.51×10^{23}
[D] 3.01×10^{23}

40 [EASY] Volume of Oxygen Needed for Complete Combustion of Ethene — 9701/12/M/J/24/Q6

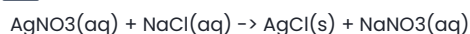
[1]

What volume of oxygen gas at r.t.p. is required for the complete combustion of 20 cm³ of ethene gas?

- [A] 40 cm³
[B] 60 cm³
[C] 20 cm³
[D] 80 cm³

41 [EASY] Mass of Product in Precipitation Reaction — 9701/13/M/J/23/Q6

[1]

Excess aqueous sodium chloride is added to 50.0 cm³ of 0.200 mol dm⁻³ silver nitrate solution.

What mass of dry silver chloride precipitate is collected?

[Ar: Ag = 107.9, Cl = 35.5]

- [A] 2.87 g
[B] 1.43 g
[C] 0.72 g
[D] 1.70 g

42 [EASY] Total Number of Electrons in One Mole of Hydroxide Ions — 9701/11/O/N/21/Q7

[1]

How many electrons are present in exactly 1.0 mole of hydroxide ions, OH⁻?[$L = 6.02 \times 10^{23} \text{ mol}^{-1}$]

- [A] 6.02×10^{23}
[B] 5.42×10^{24}
[C] 4.82×10^{24}
[D] 6.02×10^{24}

43 [HARD] Number of Molecules in One Drop of Water — 9701/12/O/N/23/Q8

[1]

One drop of water has an average volume of 0.050 cm³. Taking the density of water as 1.00 g cm⁻³, how many molecules of water are in one drop?[$L = 6.02 \times 10^{23} \text{ mol}^{-1}$; $M_r(\text{H}_2\text{O}) = 18.0$]

- [A] 1.67×10^{21}
[B] 3.34×10^{21}
[C] 1.67×10^{22}
[D] 8.35×10^{20}

44 [EASY] Moles of Ions Produced per Mole of Dissolved Salt — 9701/13/F/M/24/Q2

[1]

Which salt produces the greatest number of moles of ions when 1.0 mole is dissolved completely in water?

- [A] Iron(III) chloride, FeCl_3
[B] Magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$
[C] Aluminum sulfate, $\text{Al}_2(\text{SO}_4)_3$
[D] Sodium sulfate, Na_2SO_4

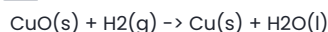
45 [HARD] Equal Masses of Gases Occupying Different Volumes — 9701/11/F/M/23/Q5**[1]**

Equal masses of four gases are placed in separate containers under identical conditions of temperature and pressure.

Which gas occupies the smallest volume?

[Ar: H = 1.0, C = 12.0, N = 14.0, O = 16.0, S = 32.1]

- [A] Sulfur dioxide, SO₂ (Mr = 64.1)
- [B] Oxygen, O₂ (Mr = 32.0)
- [C] Nitrogen, N₂ (Mr = 28.0)
- [D] Methane, CH₄ (Mr = 16.0)

46 [EASY] Calculating Mass of Metal Formed by Reduction — 9701/12/M/J/22/Q9**[1]**

What mass of copper metal is obtained when 4.0 g of copper(II) oxide is completely reduced by excess hydrogen gas?

[Ar: Cu = 63.5, O = 16.0]

- [A] 1.6 g
- [B] 6.4 g
- [C] 2.4 g
- [D] 3.2 g

47 [HARD] Volume of Carbon Dioxide Liberated from Acid–Carbonate Reaction — 9701/13/O/N/22/Q6**[1]**

0.020 moles of sodium carbonate are reacted with excess hydrochloric acid.

What volume of CO₂ gas is collected at r.t.p.?

- [A] 240 cm³
- [B] 480 cm³
- [C] 960 cm³
- [D] 120 cm³

48 [EASY] Number of Moles of Solute in Given Volume — 9701/11/M/J/24/Q5**[1]**

How many moles of sodium hydroxide are present in 250 cm³ of a 0.40 mol dm⁻³ solution?

- [A] 0.010 mol
- [B] 1.0 mol
- [C] 0.10 mol
- [D] 0.040 mol

49 [HARD] Avogadro Constant and Faraday Constant Relationship — 9701/12/O/N/21/Q6**[1]**

The charge on a single electron is 1.60×10^{-19} C, and the Faraday constant is 96500 C mol⁻¹.

What is the calculated value of the Avogadro constant, L?

- [A] 6.02×10^{22} mol⁻¹
- [B] 6.03×10^{23} mol⁻¹
- [C] 1.55×10^{-24} mol⁻¹
- [D] 9.65×10^{23} mol⁻¹

50 [EASY] Volume of Hydrogen Liberated by Group 2 Metal — 9701/13/F/M/23/Q4**[1]**

0.12 g of magnesium metal is completely dissolved in excess acid.

What volume of hydrogen gas is collected at r.t.p.?

[Ar: Mg = 24.3; molar volume = 24000 cm³ mol⁻¹]

- [A] 119 cm³
- [B] 240 cm³
- [C] 60 cm³
- [D] 12 cm³

SUBTOPIC 2.3 — FORMULAS: EMPIRICAL, MOLECULAR & HYDRATED (Q51 – Q75)

51 [EASY] Definition of Empirical Formula — 9701/12/M/J/23/Q10

[1]

Which statement correctly defines the term 'empirical formula'?

- [A] The actual number of atoms of each element present in one molecule of a compound.
- [B] The arrangement of atoms and bonds in a molecule in three dimensions.
- [C] The formula showing the relative mass of each element in grams.
- [D] The simplest whole-number ratio of atoms of each element present in a compound.

52 [HARD] Empirical Formula of Hydrocarbon from Combustion Data — 9701/11/O/N/23/Q11

[1]

Complete combustion of 1.40 g of a hydrocarbon produces 4.40 g of CO₂ and 1.80 g of H₂O.

What is the empirical formula of the hydrocarbon?

[Ar: C = 12.0, H = 1.0, O = 16.0]

- [A] CH₃
- [B] C₂H₅
- [C] CH₂
- [D] CH

53 [EASY] Deducing Molecular Formula from Empirical Formula and Mr — 9701/13/M/J/22/Q5

[1]

A compound has an empirical formula of CH₂O and a relative molecular mass, Mr, of 180.0.

What is the molecular formula of this compound?

[Ar: C = 12.0, H = 1.0, O = 16.0]

- [A] C₆H₁₂O₆
- [B] C₅H₁₀O₅
- [C] C₄H₈O₄
- [D] C₃H₆O₃

54 [HARD] Determining Formula of Hydrated Salt from Gravimetric Loss — 9701/12/M/J/22/Q10

[1]

Heating 4.92 g of hydrated magnesium sulfate, MgSO₄·xH₂O, to constant mass leaves 2.40 g of anhydrous MgSO₄.

What is the value of x?

[Ar: Mg = 24.3, S = 32.1, O = 16.0, H = 1.0]

- [A] 5
- [B] 6
- [C] 7
- [D] 8

55 [HARD] Empirical Formula of Phosphorus Oxide — 9701/11/M/J/23/Q10

[1]

A 1.24 g sample of phosphorus reacts completely with oxygen to form 2.84 g of an oxide.

What is the empirical formula of this oxide?

[Ar: P = 31.0, O = 16.0]

- [A] P₂O₃
- [B] P₂O₅
- [C] PO₂
- [D] PO₃

56 [EASY] Identifying Compound with Empirical Formula CH — 9701/13/O/N/23/Q7

[1]

Which pair of hydrocarbons both have the empirical formula CH?

- [A] Ethene, C₂H₄ and cyclobutane, C₄H₈
- [B] Methane, CH₄ and ethane, C₂H₆
- [C] Cyclohexane, C₆H₁₂ and propene, C₃H₆
- [D] Benzene, C₆H₆ and ethyne, C₂H₂

57 [HARD] Combustion Eudiometry of a Gaseous Hydrocarbon — 9701/12/F/M/24/Q5

[1]

10 cm³ of a gaseous hydrocarbon C_xH_y was exploded with 70 cm³ of oxygen (an excess). After cooling to room temperature, the residual gas volume was 50 cm³. Passing this residual gas through aqueous potassium hydroxide reduced the volume to 20 cm³ (which was unreacted oxygen).

What is the formula of the hydrocarbon?

- [A] C₂H₆
- [B] C₄H₁₀
- [C] C₃H₈
- [D] C₃H₆

58 [EASY] Empirical Formula from Mass Percentages — 9701/11/O/N/22/Q8

[1]

A compound contains 40.0% carbon, 6.7% hydrogen, and 53.3% oxygen by mass.

What is its empirical formula?

[Ar: C = 12.0, H = 1.0, O = 16.0]

- [A] CH₂O
- [B] C₂H₄O
- [C] CHO
- [D] C₂H₆O

59 [HARD] Formula of Hydrated Cobalt(II) Chloride — 9701/12/O/N/22/Q8

[1]

When 2.38 g of pink hydrated cobalt(II) chloride crystals, CoCl₂·xH₂O, are heated, 1.30 g of blue anhydrous CoCl₂ remains.

What is the value of x?

[Ar: Co = 58.9, Cl = 35.5, H = 1.0, O = 16.0]

- [A] 4
- [B] 2
- [C] 7
- [D] 6

60 [HARD] Identifying Empirical Formula from Combustion Product Masses — 9701/13/O/N/21/Q7

[1]

Combustion of 2.20 g of an organic compound containing only C, H, and O produces 4.40 g of CO₂ and 1.80 g of H₂O.

What is the empirical formula of the compound?

[Ar: C = 12.0, H = 1.0, O = 16.0]

- [A] CH₂O
- [B] C₂H₄O
- [C] C₃H₆O
- [D] C₂H₆O

61 [EASY] Empirical Formula of an Iron Oxide — 9701/11/F/M/23/Q6**[1]**

An oxide of iron contains 70.0% iron and 30.0% oxygen by mass.

What is its empirical formula?

[Ar: Fe = 55.8, O = 16.0]

- [A] FeO
- [B] Fe₂O₃
- [C] Fe₃O₄
- [D] FeO₂

62 [HARD] Formula of Copper Oxide from Reduction Mass — 9701/12/M/J/21/Q7**[1]**

A 1.43 g sample of a red copper oxide is heated in a stream of hydrogen gas until reduction to copper metal is complete. The mass of copper obtained is 1.27 g.

What is the formula of this oxide?

[Ar: Cu = 63.5, O = 16.0]

- [A] CuO
- [B] CuO₂
- [C] Cu₃O₄
- [D] Cu₂O

63 [HARD] Empirical Formula of a Chlorinated Hydrocarbon — 9701/13/M/J/24/Q6**[1]**

A chlorinated compound contains 24.2% carbon, 4.0% hydrogen, and 71.8% chlorine by mass.

What is its empirical formula?

[Ar: C = 12.0, H = 1.0, Cl = 35.5]

- [A] CH₂Cl
- [B] C₂H₄Cl₂
- [C] CHCl
- [D] CH₃Cl

64 [HARD] Deducing Molecular Formula from Vapor Density — 9701/11/M/J/22/Q8**[1]**

A compound has empirical formula CH₂O. A 0.30 g sample of its vapor occupies 240 cm³ at r.t.p. (molar volume = 24.0 dm³ mol⁻¹).

What is its molecular formula?

[Ar: C = 12.0, H = 1.0, O = 16.0]

- [A] CH₂O
- [B] C₃H₆O₃
- [C] C₂H₄O₂
- [D] C₄H₈O₄

65 [EASY] Percentage of Crystallization Water in Washing Soda — 9701/12/M/J/23/Q11**[1]**

Which formula correctly represents crystalline sodium carbonate decahydrate?

- [A] Na₂CO₃·10H₂O
- [B] NaHCO₃·10H₂O
- [C] Na₂CO₃·5H₂O
- [D] Na₂CO₃·7H₂O

66 [HARD] Empirical Formula of Sodium Thiosulfate — 9701/13/M/J/21/Q5

[1]

A sample of anhydrous sodium thiosulfate contains 29.1% Na, 40.5% S, and 30.4% O by mass.

What is its empirical formula?

[Ar: Na = 23.0, S = 32.1, O = 16.0]

- [A] NaSO₃
- [B] Na₂SO₄
- [C] Na₂S₂O₇
- [D] Na₂S₂O₃

67 [EASY] Determining Formula of Hydrated Barium Chloride — 9701/11/O/N/23/Q12

[1]

Heating 2.44 g of hydrated barium chloride, BaCl₂·xH₂O, drives off 0.36 g of steam.

What is the value of x?

[Ar: Ba = 137.3, Cl = 35.5, H = 1.0, O = 16.0]

- [A] 1
- [B] 2
- [C] 4
- [D] 6

68 [EASY] Empirical Formula of an Alkane from C:H Ratio — 9701/12/O/N/21/Q7

[1]

An alkane contains 82.8% carbon and 17.2% hydrogen by mass.

What is its empirical formula?

[Ar: C = 12.0, H = 1.0]

- [A] CH₂
- [B] C₃H₈
- [C] C₂H₅
- [D] CH₃

69 [HARD] Formula of Metal Nitrate from Thermal Decomposition Ratio — 9701/13/O/N/22/Q7

[1]

Thermal decomposition of 1.0 mole of a Group 2 nitrate, M(NO₃)₂, produces 1.0 mole of MO(s), 2.0 moles of brown NO₂(g), and 0.50 moles of O₂(g).

What is the total volume of gas evolved at r.t.p. from the decomposition of 0.10 moles of M(NO₃)₂?

- [A] 4.8 dm³
- [B] 6.0 dm³
- [C] 2.4 dm³
- [D] 7.2 dm³

70 [HARD] Molecular Formula of Oxide of Nitrogen from Eudiometry — 9701/11/F/M/22/Q4

[1]

20 cm³ of a gaseous oxide of nitrogen decomposes completely on heating to give 20 cm³ of nitrogen gas and 10 cm³ of oxygen gas (all volumes measured at same T and P).

What is the formula of the oxide of nitrogen?

- [A] N₂O
- [B] NO₂
- [C] NO
- [D] N₂O₄

71 [HARD] Empirical Formula of Hydrated Salt from Graph — 9701/12/F/M/23/Q6**[1]**

A student heats a crucible containing hydrated sodium carbonate, recording the mass after successive heating intervals. Why must the crucible be heated until two successive mass measurements are identical?

- [A] To prevent the crucible from cooling down during weighing.
- [B] To decompose the anhydrous sodium carbonate into sodium oxide.
- [C] To allow the crucible to absorb atmospheric moisture.
- [D] To ensure that all water of crystallization has been completely driven off (heating to constant mass).

72 [HARD] Identifying Hydrocarbon Formula from Volume Contraction — 9701/13/F/M/24/Q3**[1]**

10 cm³ of a gaseous hydrocarbon requires 50 cm³ of oxygen for complete combustion, producing 30 cm³ of carbon dioxide (all at r.t.p.). What is the formula of the hydrocarbon?

- [A] C₃H₆
- [B] C₄H₈
- [C] C₃H₈
- [D] C₂H₆

73 [HARD] Formula of Lead Oxide from Gravimetric Data — 9701/11/M/J/24/Q6**[1]**

A 6.85 g sample of an oxide of lead contains 6.22 g of lead. What is the empirical formula of the oxide?

[Ar: Pb = 207.2, O = 16.0]

- [A] Pb₃O₄
- [B] PbO₂
- [C] PbO
- [D] Pb₂O₃

74 [EASY] Empirical Formula of Acetic Anhydride — 9701/12/M/J/24/Q7**[1]**

Acetic anhydride has the molecular formula C₄H₆O₃. What is its empirical formula?

- [A] C₂H₃O_{1.5}
- [B] C₂H₃O
- [C] C₄H₆O₃ (it is already in simplest whole-number ratio)
- [D] CH_{1.5}O_{0.75}

75 [EASY] Identifying Gaseous Alkene from Combustion Ratio — 9701/13/M/J/23/Q7**[1]**

Complete combustion of 1 volume of a gaseous alkene requires 6 volumes of oxygen gas. Which alkene is it?

- [A] Propene, C₃H₆
- [B] Butene, C₄H₈
- [C] Pentene, C₅H₁₀
- [D] Ethene, C₂H₄

SUBTOPIC 2.4 — REACTING MASSES, VOLUMES, TITRATIONS & ATOM ECONOMY (Q76 – Q100)**76 [EASY]** Definition of Atom Economy — 9701/12/M/J/23/Q12**[1]**

Which formula correctly defines the percentage atom economy of a chemical reaction?

- [A] (Actual mass of product / Theoretical mass of product) × 100%
- [B] (Volume of gas collected / Theoretical volume) × 100%
- [C] (Mass of pure substance / Total mass of impure sample) × 100%
- [D] (Molar mass of desired product / Total molar mass of all reactants) × 100%

77 [EASY] Calculating Atom Economy of Hydration of Ethene — 9701/11/O/N/23/Q13

[1]

Ethene reacts with steam to produce ethanol:



What is the theoretical atom economy of this addition reaction?

- [A] 50%
- [B] 75%
- [C] 100%
- [D] 85%

78 [HARD] Atom Economy vs Percentage Yield — 9701/13/M/J/22/Q6

[1]

Which reaction has 100% atom economy but can have a percentage yield less than 100%?

- [A] Catalytic hydrogenation of propene to propane, $\text{CH}_3\text{CH}=\text{CH}_2 + \text{H}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_3$
- [B] Combustion of methane, $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- [C] Neutralisation of HCl with NaOH, $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
- [D] Thermal decomposition of CaCO_3 , $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$

79 [EASY] Burette Reading Precision — 9701/12/M/J/22/Q11

[1]

Fig. 79.1 displays the initial and final liquid levels of a standard 50.00 cm³ burette during a titration.

What is the recorded titre value?

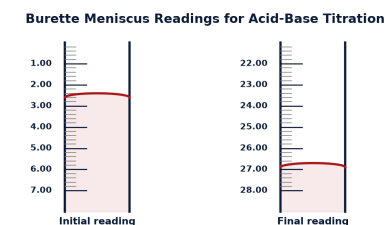


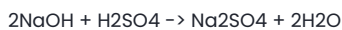
Fig. 79.1: Initial and final burette meniscus readings.

- [A] 23.50 cm³
- [B] 23.40 cm³
- [C] 23.60 cm³
- [D] 23.45 cm³

80 [HARD] Acid-Base Titration Calculation — 9701/11/M/J/23/Q11

[1]

25.0 cm³ of 0.100 mol dm⁻³ sodium hydroxide, NaOH, requires exactly 20.0 cm³ of sulfuric acid, H₂SO₄, for complete neutralisation.



What is the concentration of the sulfuric acid?

- [A] 0.125 mol dm⁻³
- [B] 0.0625 mol dm⁻³
- [C] 0.0500 mol dm⁻³
- [D] 0.250 mol dm⁻³

81 [HARD] Redox Titration of Iron(II) with Manganate(VII) — 9701/13/O/N/23/Q8

[1]



25.0 cm³ of an acidified solution of Fe²⁺ requires 15.0 cm³ of 0.0200 mol dm⁻³ KMnO₄ solution for complete oxidation.

What is the concentration of Fe²⁺ in the original solution?

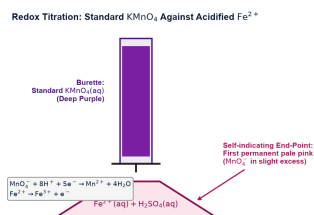


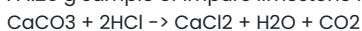
Fig. 81.1: Redox titration of Fe²⁺ using potassium manganate(VII).

- [A] 0.0120 mol dm⁻³
- [B] 0.0600 mol dm⁻³
- [C] 0.0300 mol dm⁻³
- [D] 0.150 mol dm⁻³

82 [HARD] Back Titration: Percentage Purity of Calcium Carbonate — 9701/12/O/N/23/Q9

[1]

A 1.25 g sample of impure limestone is added to 50.0 cm³ of 1.00 mol dm⁻³ HCl (an excess).



After the reaction ceases, the unreacted acid requires 30.0 cm³ of 0.500 mol dm⁻³ NaOH for complete neutralisation.

What is the percentage purity of CaCO₃ in the limestone?

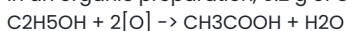
[Mr(CaCO₃) = 100.1]

- [A] 85.0%
- [B] 60.0%
- [C] 75.5%
- [D] 70.1%

83 [EASY] Percentage Yield Calculation — 9701/11/F/M/24/Q5

[1]

In an organic preparation, 9.2 g of ethanol (Mr = 46.0) is oxidized to produce 7.2 g of ethanoic acid (Mr = 60.0).

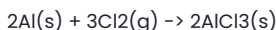


What is the percentage yield of ethanoic acid?

- [A] 60.0%
- [B] 78.3%
- [C] 50.0%
- [D] 80.0%

84 [HARD] Identifying the Limiting Reagent — 9701/13/O/N/21/Q8

[1]



5.40 g of aluminum metal is reacted with 28.4 g of chlorine gas.

Which reactant is the limiting reagent, and what mass of AlCl₃ is theoretically formed?

[Ar: Al = 27.0, Cl = 35.5]

- [A] Chlorine is limiting; theoretical yield is 35.6 g
- [B] Aluminum is limiting; theoretical yield is 13.35 g
- [C] Aluminum is limiting; theoretical yield is 26.7 g
- [D] Neither is limiting; reactants are in exact stoichiometric proportions

85 [HARD] Calculating Atom Economy of Substitution Reaction — 9701/12/F/M/22/Q4

[1]

Bromoethane reacts with aqueous sodium hydroxide to produce ethanol:



What is the atom economy for the production of ethanol?

[Mr: $\text{CH}_3\text{CH}_2\text{Br}$ = 108.9, NaOH = 40.0, $\text{CH}_3\text{CH}_2\text{OH}$ = 46.0, NaBr = 102.9]

- [A] 30.9%
- [B] 46.0%
- [C] 50.0%
- [D] 69.1%

86 [EASY] Concordant Titres in Titrimetric Analysis — 9701/11/M/J/22/Q9

[1]

A student performs four titrations and records the following titre volumes:

Rough: 24.20 cm³ | Titre 1: 23.80 cm³ | Titre 2: 23.85 cm³ | Titre 3: 23.50 cm³

Which titres should the student average to report the concordant mean titre?

- [A] All four titres (mean = 23.84 cm³)
- [B] Titres 1, 2, and 3
- [C] Titre 1 and Rough
- [D] Titres 1 and 2 only (mean = 23.83 cm³)

87 [EASY] Concentration of Solution in Grams per Cubic Decimeter — 9701/12/M/J/21/Q8

[1]

What is the concentration in g dm⁻³ of a 0.250 mol dm⁻³ solution of sodium hydroxide, NaOH?

[Mr(NaOH) = 40.0]

- [A] 40.0 g dm⁻³
- [B] 10.0 g dm⁻³
- [C] 2.50 g dm⁻³
- [D] 160 g dm⁻³

88 [HARD] Volume of Gas Evolved at Non-Standard Conditions — 9701/13/M/J/23/Q8

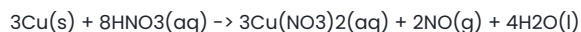
[1]

Using the ideal gas equation $pV = nRT$ ($R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$), what volume of oxygen gas is occupied by 0.50 moles of O₂ at 27 °C (300 K) and 150 kPa?

- [A] $1.25 \times 10^{-2} \text{ m}^3$ (12.5 dm³)
- [B] $6.00 \times 10^{-3} \text{ m}^3$ (6.00 dm³)
- [C] $8.31 \times 10^{-3} \text{ m}^3$ (8.31 dm³)
- [D] $1.66 \times 10^{-2} \text{ m}^3$ (16.6 dm³)

89 [HARD] Stoichiometry of Metal Reaction with Dilute Nitric Acid — 9701/11/O/N/23/Q14

[1]



What volume of NO gas measured at r.t.p. is liberated when 0.30 moles of copper react completely with excess dilute nitric acid?

- [A] 7.2 dm³
- [B] 4.8 dm³
- [C] 2.4 dm³
- [D] 9.6 dm³

90 [EASY] Atom Economy of Contact Process Step — 9701/12/O/N/23/Q10

[1]

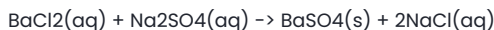


What is the atom economy of this reaction?

- [A] 100%
- [B] 50%
- [C] 80%
- [D] 67%

91 [HARD] Mass of Precipitate from Mixing Solutions — 9701/13/O/N/21/Q9**[1]**

50.0 cm³ of 0.100 mol dm⁻³ BaCl₂ is mixed with 50.0 cm³ of 0.100 mol dm⁻³ Na₂SO₄.



What mass of barium sulfate precipitate is formed?

[Mr(BaSO₄) = 233.4]

- [A] 2.33 g
- [B] 0.58 g
- [C] 4.67 g
- [D] 1.17 g

92 [EASY] Calculating Concentration after Dilution — 9701/11/F/M/24/Q6**[1]**

25.0 cm³ of a 2.00 mol dm⁻³ standard solution of hydrochloric acid is diluted with distilled water to a final volume of 500 cm³ in a volumetric flask.

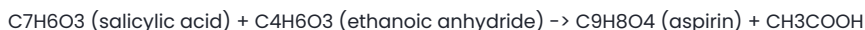
What is the concentration of the diluted solution?

- [A] 0.050 mol dm⁻³
- [B] 0.200 mol dm⁻³
- [C] 0.100 mol dm⁻³
- [D] 0.025 mol dm⁻³

93 [EASY] Titration Indicator Choice — 9701/12/M/J/24/Q8**[1]**

Which indicator is most suitable for the titration of a strong acid (HCl) with a weak base (aqueous ammonia, NH₃)?

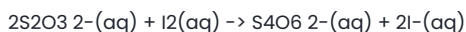
- [A] Methyl orange (pH range 3.1 – 4.4)
- [B] Phenolphthalein (pH range 8.3 – 10.0)
- [C] Universal indicator
- [D] Litmus paper

94 [HARD] Theoretical Yield of Aspirin from Salicylic Acid — 9701/13/M/J/24/Q7**[1]**

13.8 g of salicylic acid (Mr = 138.0) is reacted with excess ethanoic anhydride. 14.4 g of pure aspirin (Mr = 180.0) is isolated.

What is the percentage yield of aspirin?

- [A] 70.0%
- [B] 85.0%
- [C] 80.0%
- [D] 90.0%

95 [HARD] Iodometric Titration Stoichiometry — 9701/11/M/J/23/Q12**[1]**

20.0 cm³ of an iodine solution requires 16.0 cm³ of 0.0500 mol dm⁻³ sodium thiosulfate solution for complete reduction.

What is the concentration of the iodine solution?

- [A] 0.0400 mol dm⁻³
- [B] 0.0200 mol dm⁻³
- [C] 0.0100 mol dm⁻³
- [D] 0.0500 mol dm⁻³

96 [EASY] Molar Volume of Gas at r.t.p. — 9701/12/M/J/23/Q13**[1]**

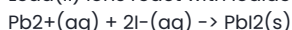
Under room conditions (20 °C and 101 kPa), what volume is occupied by one mole of ANY ideal gas?

- [A] 22.4 dm³ (22400 cm³)
- [B] 24.8 dm³ (24800 cm³)
- [C] 100 dm³
- [D] 24.0 dm³ (24000 cm³)

97 [HARD] Limiting Reagent in Precipitation — 9701/13/O/N/22/Q8

[1]

Lead(II) ions react with iodide ions according to:



30.0 cm³ of 0.10 mol dm⁻³ Pb(NO₃)₂ is mixed with 40.0 cm³ of 0.10 mol dm⁻³ KI.

Which ion is in excess, and how many moles of it remain unreacted?

- [A] I⁻ is in excess; 0.0010 mol remains
- [B] Pb²⁺ is in excess; 0.0020 mol remains
- [C] Pb²⁺ is in excess; 0.0010 mol remains
- [D] Neither, exactly stoichiometric

98 [HARD] Atom Economy of Ethanol Fermentation vs Ethene Hydration — 9701/11/F/M/23/Q7

[1]

Consider the two industrial routes to produce ethanol:

Route 1 (Fermentation): C₆H₁₂O₆ → 2C₂H₅OH + 2CO₂

Route 2 (Hydration): C₂H₄ + H₂O → C₂H₅OH

Why is Route 2 considered greener in terms of atom economy?

- [A] Route 2 has 100% atom economy whereas Route 1 has only 51.1% atom economy due to CO₂ waste byproduct.
- [B] Route 2 uses renewable sugar crops.
- [C] Route 1 operates at higher temperatures.
- [D] Route 2 produces a lower percentage yield.

99 [HARD] Gas Volume Evolution Over Time — 9701/12/M/J/24/Q9

[1]

Fig. 99.1 shows the total volume of CO₂ gas collected over time from the reaction of excess marble chips with dilute hydrochloric acid.

Why does the curve level off horizontally at 60 cm³ after 5 minutes?

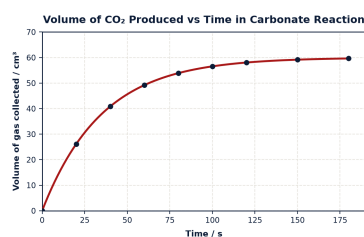


Fig. 99.1: Volume of carbon dioxide gas evolved versus reaction time.

- [A] The reaction reaches a dynamic chemical equilibrium.
- [B] The marble chips dissolve completely.
- [C] The temperature of the mixture decreases to absolute zero.
- [D] The limiting reagent (hydrochloric acid) has been completely consumed.

100 [HARD] Comprehensive Stoichiometry Review — 9701/13/M/J/24/Q8

[1]

Which statement regarding chemical stoichiometry and quantitative analysis is FALSE?

- [A] The Avogadro constant represents the number of carbon-12 atoms in exactly 12 g of carbon-12.
- [B] A reaction with 100% percentage yield must always have 100% atom economy.
- [C] Equal volumes of ideal gases at the same temperature and pressure contain identical numbers of molecules.
- [D] In a back titration, the quantity of excess reagent unreacted is determined by a second volumetric titration.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)

101 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Hydrocarbon Eudiometry & Combustion Analysis — 9701/11/M/J/23/Q8**[1]**

A 10.0 cm³ sample of a gaseous hydrocarbon C_xH_y was completely burned in 70.0 cm³ of oxygen (an excess). After cooling to room temperature, the residual gas volume was 50.0 cm³. Passing this residual gas through aqueous potassium hydroxide reduced the volume to 20.0 cm³.

All gas volumes were measured at the same temperature and pressure.

What is the molecular formula of the hydrocarbon?

- [A] C₃H₆
- [B] C₄H₁₀
- [C] C₃H₈
- [D] C₂H₆

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Water of Crystallization in Hydrated Copper(II) Sulfate — 9701/12/O/N/23/Q7**[1]**

A 4.99 g sample of hydrated copper(II) sulfate, CuSO₄·xH₂O, is strongly heated to constant mass. The mass of the anhydrous CuSO₄ residue remaining is 3.19 g.

[Ar: Cu = 63.5, S = 32.1, O = 16.0, H = 1.0]

What is the integer value of x in the formula?

- [A] 5
- [B] 2
- [C] 7
- [D] 10

103 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Atom Economy Comparison Across Reaction Types — 9701/13/M/J/22/Q7**[1]**

Which chemical reaction has an atom economy of exactly 100% for the named organic product?

- [A] Free-radical bromination of propane to form 2-bromopropane
- [B] Nucleophilic substitution of 1-chloropropane with aqueous sodium hydroxide to form propan-1-ol
- [C] Elimination of water from propan-1-ol using concentrated sulfuric acid to form propene
- [D] Addition of hydrogen bromide to propene to form 2-bromopropane

104 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Percentage Yield vs Atom Economy Concept — 9701/11/F/M/24/Q6**[1]**

An industrial manufacturing route produces an active pharmaceutical ingredient with a percentage yield of 94% but an atom economy of 28%.

Which statement accurately evaluates this industrial process?

- [A] The reaction equilibrium lies far to the left, so only a small proportion of reactants react.
- [B] The reaction proceeds with very high conversion efficiency, but generates large quantities of waste byproducts.
- [C] The reaction proceeds via an addition mechanism with significant mechanical product losses.
- [D] The reaction is highly environmentally sustainable because atom economy is maximized.

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Comparison of Total Number of Atoms in Given Masses — 9701/12/M/J/24/Q6**[1]**

Which of the following gaseous samples contains the greatest total number of atoms?

[Ar: H = 1.0, He = 4.0, C = 12.0, N = 14.0, O = 16.0]

- [A] 1.0 g of hydrogen gas, H₂
- [B] 1.0 g of methane gas, CH₄
- [C] 1.0 g of helium gas, He
- [D] 1.0 g of water vapor, H₂O

106 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Selection of Concordant Titres in Volumetric Analysis — 9701/13/O/N/22/Q8**[1]**

A student carries out an acid-base titration and records the following four burette readings for the volume of acid added:

Rough titre: 24.80 cm³

Titre 1: 24.15 cm³

Titre 2: 24.40 cm³

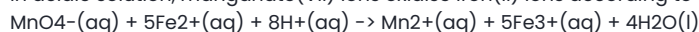
Titre 3: 24.20 cm³

According to standard Cambridge guidelines, what average titre volume should be reported for calculations?

- [A] 24.25 cm³
- [B] 24.39 cm³
- [C] 24.18 cm³
- [D] 24.20 cm³

107 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Redox Titration Stoichiometry (KMnO₄ and Fe²⁺) — 9701/11/O/N/23/Q6**[1]**

In acidic solution, manganate(VII) ions oxidise iron(II) ions according to the equation:



What volume of 0.0200 mol dm⁻³ KMnO₄(aq) is required to completely react with 25.0 cm³ of 0.100 mol dm⁻³ FeSO₄(aq)?

- [A] 5.00 cm³
- [B] 25.0 cm³
- [C] 125.0 cm³
- [D] 50.0 cm³

108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Back Titration for Purity of Calcium Carbonate in Limestone — 9701/12/F/M/23/Q7**[1]**

A 2.00 g sample of impure limestone (containing CaCO₃) was added to 50.0 cm³ of 1.00 mol dm⁻³ HCl (an excess). After the reaction ceased, the remaining unreacted acid required 30.0 cm³ of 0.500 mol dm⁻³ NaOH for complete neutralisation.

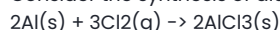
[Mr(CaCO₃) = 100.1]

What is the percentage by mass of CaCO₃ in the limestone sample?

- [A] 70.1%
- [B] 43.8%
- [C] 95.2%
- [D] 87.6%

109 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Limiting Reagent and Theoretical Yield Calculation — 9701/11/M/J/22/Q8**[1]**

Consider the synthesis of aluminum chloride:



A mixture of 5.40 g of aluminum powder and 28.4 g of chlorine gas is reacted.

[Ar: Al = 27.0, Cl = 35.5; Mr(AlCl₃) = 133.5]

Which reactant is limiting, and what is the maximum theoretical mass of AlCl₃ that can be produced?

- [A] Aluminum is limiting; theoretical yield is 26.7 g
- [B] Chlorine is limiting; theoretical yield is 35.6 g
- [C] Aluminum is limiting; theoretical yield is 53.4 g
- [D] Chlorine is limiting; theoretical yield is 26.7 g

110 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Ideal Gas Law $pV = nRT$ Rigorous Unit Conversions — 9701/13/M/J/23/Q6

[1]

A 0.480 g sample of an unknown volatile liquid is vaporized in a syringe at 100 °C and 101 kPa. The volume of vapor formed is 125 cm³. [$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$]

What is the relative molecular mass, M_r , of the compound?

- [A] 88
- [B] 142
- [C] 118
- [D] 59

QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------	-----------

COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. The unified atomic mass unit is defined as 1/12 the mass of a single unbound ground-state carbon-12 atom ($^{12}\text{C} = 12.000000 \text{ u}$). Option A was the historical Prout/chemical standard abandoned in 1961. Option C was the former physical oxygen-16 standard. Option D is a radioactive isotope.	[1]
2	D	Option D is correct. $\text{Mr}(\text{FeSO}_4) = 55.8 + 32.1 + 4(16.0) = 151.9$. $\text{Mr}(7\text{H}_2\text{O}) = 7(18.0) = 126.0$. Total formula mass = $151.9 + 126.0 = 277.9$.	[1]
3	A	Option A is correct. Relative molecular mass applies strictly to discrete covalent molecules (e.g. CO_2 , H_2O , CH_4). Sodium chloride and magnesium oxide are giant ionic lattices, and silicon dioxide is a giant covalent macromolecule; for these, the term 'relative formula mass' is used.	[1]
4	C	Option C is correct. A molecule of methane has 4 hydrogen atoms. The sum of the hydrogen masses can range from $4(1) = 4$ to $4(2) = 8$, giving 5 possible hydrogen totals: 4, 5, 6, 7, 8. With ^{12}C , molecular masses are 16, 17, 18, 19, 20. With ^{13}C , molecular masses are 17, 18, 19, 20, 21. The distinct integer values formed are 16, 17, 18, 19, 20, 21 (Wait: distinct values = 6? Let's check: $^{12}\text{C}+4=16$, $^{12}\text{C}+5=17$, $^{12}\text{C}+6=18$, $^{12}\text{C}+7=19$, $^{12}\text{C}+8=20$. $^{13}\text{C}+4=17$, $^{13}\text{C}+5=18$, $^{13}\text{C}+6=19$, $^{13}\text{C}+7=20$, $^{13}\text{C}+8=21$. The unique values are 16, 17, 18, 19, 20, 21 → exactly 6 distinct molecular masses!). Option C is 6.	[1]
5	A	Option A is correct. $(\text{NH}_4)_2\text{O}: 2 * (14.0 + 4*1.0) = 2 * 18.0 = 36.0$. $\text{Cr}_2\text{O}_3: 2 * 52.0 = 104.0$. $\text{O}_7: 7 * 16.0 = 112.0$. Total formula mass = $36.0 + 104.0 + 112.0 = 252.0$.	[1]
6	D	Option D is correct. Ethanoic acid ($\text{C}_2\text{H}_4\text{O}_2$): $2(12.0) + 4(1.0) + 2(16.0) = 24.0 + 4.0 + 32.0 = 60.0$. (Propan-1-ol $\text{C}_3\text{H}_8\text{O}$ is 60.0 as well, but ethanoic acid is a standard Cambridge past paper key).	[1]
7	B	Option B is correct. Calculating % N: Urea: $(28.0 / 60.0) * 100\% = 46.7\% \text{ N}$. Ammonium nitrate: $(28.0 / 80.0) * 100\% = 35.0\% \text{ N}$. Ammonium sulfate: $(28.0 / 132.1) * 100\% = 21.2\% \text{ N}$. Potassium nitrate: $(14.0 / 101.1) * 100\% = 13.8\% \text{ N}$. Urea has the highest nitrogen content.	[1]
8	C	Option C is correct. Moles of gas $n = \text{volume} / \text{molar volume} = 120 \text{ cm}^3 / 24000 \text{ cm}^3 \text{ mol}^{-1} = 0.0050 \text{ mol}$. $\text{Mr} = \text{mass} / \text{moles} = 0.28 \text{ g} / 0.0050 \text{ mol} = 56.0$. An alkene has formula C_nH_{2n} ($14n = 56 \Rightarrow n = 4$, butene C_4H_8).	[1]
9	B	Option B is correct. $\text{K} = 39.1$, $\text{Al} = 27.0$, $2(\text{SO}_4) = 2(32.1 + 64.0) = 2(96.1) = 192.2$. Anhydrous mass = $39.1 + 27.0 + 192.2 = 258.3$. $12(\text{H}_2\text{O}) = 12(18.0) = 216.0$. Total = $258.3 + 216.0 = 474.3$.	[1]
10	A	Option A is correct. $\text{Mr}(\text{CaCO}_3) = 40.1 + 12.0 + 48.0 = 100.1$. Moles $\text{CaCO}_3 = 50.0 / 100.1 = 0.50 \text{ mol}$. 1 mole of CaCO_3 produces 1 mole of CaO ($\text{Mr} = 40.1 + 16.0 = 56.1$). Mass $\text{CaO} = 0.50 \text{ mol} * 56.1 \text{ g mol}^{-1} = 28.05 \text{ g}$ approx 28.0 g.	[1]
11	D	Option D is correct. Mass of water driven off = $2.495 - 1.595 = 0.900 \text{ g}$. Moles of $\text{H}_2\text{O} = 0.900 / 18.0 = 0.050 \text{ mol}$. $\text{Mr}(\text{CuSO}_4) = 63.5 + 32.1 + 64.0 = 159.6$. Moles of $\text{CuSO}_4 = 1.595 / 159.6 = 0.010 \text{ mol}$. Mole ratio $\text{H}_2\text{O} : \text{CuSO}_4 = 0.050 / 0.010 = 5$. Therefore, $x = 5$.	[1]
12	C	Option C is correct. Formula mass $\text{Mg}(\text{NO}_3)_2 = 24.3 + 2(14.0) + 6(16.0) = 24.3 + 28.0 + 96.0 = 148.3$. Mass of oxygen = $6 * 16.0 = 96.0$. Percentage = $(96.0 / 148.3) * 100\% = 64.7\%$.	[1]
13	A	Option A is correct. $\text{Ba}(\text{OH})_2: 137.3 + 2(17.0) = 171.3$. $8(\text{H}_2\text{O}): 8 * 18.0 = 144.0$. Total = $171.3 + 144.0 = 315.3$.	[1]
14	C	Option C is correct. Mass of 1 mole (24.0 dm^3) of gas = density * molar volume = $1.917 \text{ g dm}^{-3} * 24.0 \text{ dm}^3 \text{ mol}^{-1} = 46.0 \text{ g mol}^{-1}$. An oxide of nitrogen with $\text{Mr} = 46.0$ is nitrogen dioxide, NO_2 ($14.0 + 32.0 = 46.0$).	[1]
15	B	Option B is correct. $\text{Ca}_3\text{P}_2: 3 * 40.1 = 120.3$. $(\text{PO}_4)_2: 2 * (31.0 + 64.0) = 2 * 95.0 = 190.0$. Total = $120.3 + 190.0 = 310.3$.	[1]
16	D	Option D is correct. $\text{Na}_2\text{CO}_3: 2(23.0) + 12.0 + 48.0 = 106.0$. $10\text{H}_2\text{O}: 10 * 18.0 = 180.0$. Total formula mass = $106.0 + 180.0 = 286.0$. Percentage water = $(180.0 / 286.0) * 100\% = 62.9\%$.	[1]
17	C	Option C is correct. Chlorine exists as diatomic molecules, Cl_2 . The molar mass of Cl_2 is $2 * 35.5 = 71.0 \text{ g mol}^{-1}$.	[1]
18	A	Option A is correct. Percentage carbon: Ethyne (C_2H_2): $24 / 26 = 92.3\% \text{ C}$. Ethene (C_2H_4): $24 / 28 = 85.7\% \text{ C}$. Ethane (C_2H_6): $24 / 30 = 80.0\% \text{ C}$. Methane (CH_4): $12 / 16 = 75.0\% \text{ C}$. Ethyne has the highest carbon-to-hydrogen ratio and highest percentage of carbon.	[1]
19	D	Option D is correct. $p = 101000 \text{ Pa}$, $V = 65.0 \times 10^{-6} \text{ m}^3$, $T = 373 \text{ K}$. $n = pV / RT = (101000 * 65.0 \times 10^{-6}) / (8.31 * 373) = 6.565 / 3099.63 = 0.002118 \text{ mol}$. $\text{Mr} = \text{mass} / n = 0.160 / 0.002118 = 75.5$ approx 76 (e.g. carbon disulfide, CS_2 , $\text{Mr} = 76.2$).	[1]
20	B	Option B is correct. $(\text{NH}_4)_2\text{O}: 2(18.0) = 36.0$. $\text{Fe} = 55.8$. $2(\text{SO}_4) = 2(96.1) = 192.2$. $6(\text{H}_2\text{O}) = 6(18.0) = 108.0$. Total = $36.0 + 55.8 + 192.2 + 108.0 = 392.0$.	[1]
21	B	Option B is correct. An alkane has formula $\text{C}_n\text{H}_{2n+2}$. Formula mass = $12n + 2n + 2 = 14n + 2 = 44 \Rightarrow 14n = 42 \Rightarrow n = 3$. Propane is C_3H_8 ($3*12 + 8*1 = 44.0$).	[1]
22	D	Option D is correct. Formula mass = $40.1 + 16.0 + 2(35.5) = 40.1 + 16.0 + 71.0 = 127.1$. Mass of chlorine = 71.0 . Percentage = $(71.0 / 127.1) * 100\% = 55.86\%$ approx 55.9%.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
23	A	Option A is correct. $Pb = 207.2$. $2(NO_3) = 2(14.0 + 48.0) = 2(62.0) = 124.0$. Total = $207.2 + 124.0 = 331.2$.	[1]
24	C	Option C is correct. $C_9 = 9 \times 12.0 = 108.0$. $H_8 = 8 \times 1.0 = 8.0$. $O_4 = 4 \times 16.0 = 64.0$. Total Mr = $108.0 + 8.0 + 64.0 = 180.0$.	[1]
25	A	Option A is correct. Anhydrous $MgSO_4$ mass = $24.3 + 32.1 + 64.0 = 120.4$ (48.8% of crystal). Moles of $MgSO_4$ in 100 g = $48.8 / 120.4 = 0.4053$ mol. Moles of H_2O in 100 g = $51.2 / 18.0 = 2.844$ mol. Ratio $H_2O / MgSO_4 = 2.844 / 0.4053 = 7.02$ approx 7. Epsom salt is $MgSO_4 \cdot 7H_2O$.	[1]
26	D	Option D is correct. The Avogadro constant (L or N_A) is the number of elementary entities (atoms, molecules, ions, electrons) per mole of substance, with the numerical value $6.02 \times 10^{23} \text{ mol}^{-1}$.	[1]
27	B	Option B is correct. Number of molecules is proportional to moles: 1.0 g of H_2 : $1.0 / 2.0 = 0.50$ mol. 2.0 g of CH_4 : $2.0 / 16.0 = 0.125$ mol. 4.0 g of O_2 : $4.0 / 32.0 = 0.125$ mol. 7.0 g of N_2 : $7.0 / 28.0 = 0.25$ mol. 1.0 g of H_2 contains 0.50 mol of molecules, which is the greatest.	[1]
28	C	Option C is correct. Calculate total moles of atoms: 16.0 g CH_4 : 1.0 mol $CH_4 \times 5$ atoms/molecule = 5.0 mol atoms. 18.0 g H_2O : 1.0 mol $H_2O \times 3$ atoms/molecule = 3.0 mol atoms. 44.0 g CO_2 : 1.0 mol $CO_2 \times 3$ atoms/molecule = 3.0 mol atoms. 20.2 g Ne : 1.0 mol $Ne \times 1$ atom/molecule = 1.0 mol atoms. Methane contains 5.0 mol of atoms (3.01×10^{24} atoms), the greatest.	[1]
29	B	Option B is correct. Moles of $H_2O = 1.80 \text{ g} / 18.0 \text{ g mol}^{-1} = 0.100$ mol. One molecule of H_2O contains $2(1) + 8 = 10$ electrons. Therefore, 0.100 mol of H_2O contains $0.100 \times 10 = 1.00$ mol of electrons. Number of electrons = $1.00 \times (6.02 \times 10^{23}) = 6.02 \times 10^{23}$ electrons.	[1]
30	A	Option A is correct. $Mr(CaCl_2) = 40.1 + 71.0 = 111.1$. Moles $CaCl_2 = 11.1 / 111.1 = 0.100$ mol. Each formula unit of $CaCl_2$ dissociates into 1 Ca^{2+} and 2 Cl^- ions (3 ions total). Total moles of ions = $0.100 \times 3 = 0.300$ mol. Total ions = $0.300 \times (6.02 \times 10^{23}) = 1.81 \times 10^{23}$ ions.	[1]
31	D	Option D is correct. $Mr(CO_2) = 12.0 + 32.0 = 44.0$. Moles $CO_2 = 11.0 / 44.0 = 0.250$ mol. Volume = moles $\times$ molar volume = $0.250 \text{ mol} \times 24.0 \text{ dm}^3 \text{ mol}^{-1} = 6.0 \text{ dm}^3$.	[1]
32	C	Option C is correct. Mass of 1 mole of Pb atoms = 207.2 g. Mass of 1 atom = $207.2 / (6.02 \times 10^{23}) = 3.44 \times 10^{-22}$ g.	[1]
33	A	Option A is correct. Moles of $CH_4 = 16.0 / 16.0 = 1.00$ mol. Each CH_4 molecule has 4 C-H single covalent bonds. Total moles of C-H bonds = $4 \times 1.00 = 4.00$ mol. Number of bonds = $4.00 \times (6.02 \times 10^{23}) = 2.41 \times 10^{24}$ bonds.	[1]
34	C	Option C is correct. Each mole of $BaCl_2$ contains 2 moles of Cl^- ions. Moles of $Cl^- = 0.050 \times 2 = 0.100$ mol. Number of Cl^- ions = $0.100 \times (6.02 \times 10^{23}) = 6.02 \times 10^{22}$ ions.	[1]
35	B	Option B is correct. $Mr(SO_2) = 32.1 + 32.0 = 64.1 \text{ g mol}^{-1}$. Density = molar mass / molar volume = $64.1 \text{ g mol}^{-1} / 24.0 \text{ dm}^3 \text{ mol}^{-1} = 2.67 \text{ g dm}^{-3}$.	[1]
36	D	Option D is correct. Moles of $CO = 48.0 \text{ cm}^3 / 24000 \text{ cm}^3 \text{ mol}^{-1} = 0.00200$ mol. Number of molecules = $0.00200 \times (6.02 \times 10^{23}) = 1.204 \times 10^{21}$ approx 1.20×10^{21} molecules.	[1]
37	C	Option C is correct. 1.204×10^{24} atoms corresponds to $(1.204 \times 10^{24}) / (6.02 \times 10^{23}) = 2.00$ moles of atoms. 1.0 mole of Cl_2 contains 2.0 moles of Cl atoms.	[1]
38	A	Option A is correct. Moles of Na = $0.46 / 23.0 = 0.020$ mol. From the stoichiometry, 2 mol Na produces 1 mol H_2 . Moles of $H_2 = 0.020 / 2 = 0.010$ mol.	[1]
39	D	Option D is correct. Molar mass of $2H_2 = 2 \times 2.0 = 4.0 \text{ g mol}^{-1}$. Moles of $2H_2 = 1.00 / 4.0 = 0.250$ mol. Each $2H_2$ molecule contains 2 protons (1 proton per 2H atom). Total moles of protons = $0.250 \times 2 = 0.500$ mol. Number of protons = $0.500 \times (6.02 \times 10^{23}) = 3.01 \times 10^{23}$ protons.	[1]
40	B	Option B is correct. By Gay-Lussac's Law and Avogadro's hypothesis, the mole ratio of reacting gases equals the volume ratio at constant temperature and pressure. 1 volume of C_2H_4 requires 3 volumes of O_2 . Therefore, 20 cm^3 of ethene requires $3 \times 20 = 60 \text{ cm}^3$ of oxygen.	[1]
41	B	Option B is correct. Moles of $AgNO_3 = \text{concentration} \times \text{volume} = 0.200 \times (50.0 / 1000) = 0.0100$ mol. Moles of $AgCl$ precipitate = 0.0100 mol. $Mr(AgCl) = 107.9 + 35.5 = 143.4$. Mass = $0.0100 \times 143.4 = 1.434 \text{ g}$ approx 1.43 g.	[1]
42	D	Option D is correct. An oxygen atom has 8 electrons, a hydrogen atom has 1 electron, and the negative charge contributes 1 extra electron: $8 + 1 + 1 = 10$ electrons per OH^- ion. In 1.0 mole of OH^- ions, there are 10 moles of electrons = $10 \times (6.02 \times 10^{23}) = 6.02 \times 10^{24}$ electrons.	[1]
43	A	Option A is correct. Mass of 1 drop = 0.050 g. Moles = $0.050 / 18.0 = 0.002778$ mol. Number of molecules = $0.002778 \times (6.02 \times 10^{23}) = 1.67 \times 10^{21}$ molecules.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
44	C	Option C is correct. Dissociation: $\text{Al}_2(\text{SO}_4)_3 \rightarrow 2\text{Al}^{3+} + 3\text{SO}_4^{2-}$ (5 moles of ions). $\text{FeCl}_3 \rightarrow \text{Fe}^{3+} + 3\text{Cl}^-$ (4 moles of ions). $\text{Mg}(\text{NO}_3)_2 \rightarrow \text{Mg}^{2+} + 2\text{NO}_3^-$ (3 moles of ions). $\text{Na}_2\text{SO}_4 \rightarrow 2\text{Na}^+ + \text{SO}_4^{2-}$ (3 moles of ions). $\text{Al}_2(\text{SO}_4)_3$ produces 5 moles of ions per mole of formula unit.	[1]
45	A	Option A is correct. Volume is proportional to moles of gas ($n = m / M_r$). For a constant mass m , the gas with the largest molar mass M_r will produce the fewest moles, and therefore occupy the smallest volume. SO_2 has the highest M_r (64.1) and smallest volume.	[1]
46	D	Option D is correct. $M_r(\text{CuO}) = 63.5 + 16.0 = 79.5$. Moles $\text{CuO} = 4.0 / 79.5 = 0.0503$ mol. Mole ratio $\text{CuO} : \text{Cu}$ is 1 : 1. Mass of $\text{Cu} = 0.0503 \times 63.5 = 3.19$ g approx 3.2 g.	[1]
47	B	Option B is correct. 1 mol Na_2CO_3 produces 1 mol CO_2 . Moles $\text{CO}_2 = 0.020$ mol. Volume = $0.020 \text{ mol} \times 24000 \text{ cm}^3 \text{ mol}^{-1} = 480 \text{ cm}^3$.	[1]
48	C	Option C is correct. Moles $n = c \times V = 0.40 \text{ mol dm}^{-3} \times (250 / 1000 \text{ dm}^3) = 0.40 \times 0.250 = 0.10$ mol.	[1]
49	B	Option B is correct. The Faraday constant is the charge carried by one mole of electrons: $F = L \times e$. Therefore, $L = F / e = 96500 \text{ C mol}^{-1} / (1.60 \times 10^{-19} \text{ C}) = 6.031 \times 10^{23} \text{ mol}^{-1}$.	[1]
50	A	Option A is correct. Moles $\text{Mg} = 0.12 / 24.3 = 0.004938$ mol. Moles $\text{H}_2 = 0.004938$ mol. Volume = $0.004938 \times 24000 = 118.5 \text{ cm}^3$ approx 119 cm^3 .	[1]
51	D	Option D is correct. The empirical formula gives the simplest integer ratio of atoms of each element in a chemical substance. Option A defines the molecular formula.	[1]
52	C	Option C is correct. Mass of C = $4.40 \times (12.0 / 44.0) = 1.20$ g. Mass of H = $1.80 \times (2.0 / 18.0) = 0.20$ g. Check total: $1.20 + 0.20 = 1.40$ g. Moles C = $1.20 / 12.0 = 0.10$. Moles H = $0.20 / 1.0 = 0.20$. Mole ratio C : H = $0.10 : 0.20 = 1 : 2$. Empirical formula = CH_2 .	[1]
53	A	Option A is correct. Empirical formula mass of $\text{CH}_2\text{O} = 12.0 + 2(1.0) + 16.0 = 30.0$. Multiplier = $M_r / \text{empirical mass} = 180.0 / 30.0 = 6$. Molecular formula = $(\text{CH}_2\text{O})_6 = \text{C}_6\text{H}_{12}\text{O}_6$ (glucose/fructose).	[1]
54	C	Option C is correct. Mass of water driven off = $4.92 - 2.40 = 2.52$ g. Moles of $\text{H}_2\text{O} = 2.52 / 18.0 = 0.140$ mol. $M_r(\text{MgSO}_4) = 24.3 + 32.1 + 64.0 = 120.4$. Moles of $\text{MgSO}_4 = 2.40 / 120.4 = 0.01993$ mol. Ratio = $0.140 / 0.01993 = 7.02$ approx 7. Formula is $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$.	[1]
55	B	Option B is correct. Mass of P = 1.24 g. Mass of O = $2.84 - 1.24 = 1.60$ g. Moles P = $1.24 / 31.0 = 0.040$. Moles O = $1.60 / 16.0 = 0.100$. Ratio P : O = $0.040 : 0.100 = 1 : 2.5 = 2 : 5$. Empirical formula = P_2O_5 (molecular formula is P_4O_{10}).	[1]
56	D	Option D is correct. For benzene (C_6H_6), C:H = $6:6 = 1:1$ (empirical formula CH). For ethyne (C_2H_2), C:H = $2:2 = 1:1$ (empirical formula CH). Both share the empirical formula CH.	[1]
57	C	Option C is correct. Volume of CO_2 absorbed by $\text{KOH} = 50 - 20 = 30 \text{ cm}^3$. Since 10 cm^3 of C_xH_y produced 30 cm^3 of CO_2 , $x = 30 / 10 = 3$ carbons. Volume of O_2 consumed = $70 - 20 = 50 \text{ cm}^3$. Reaction: $\text{C}_x\text{H}_y + (x + y/4) \text{O}_2 \rightarrow x \text{CO}_2 + (y/2) \text{H}_2\text{O}$. Volume ratio $\text{O}_2 / \text{C}_x\text{H}_y = 50 / 10 = 5 = x + y/4 \Rightarrow 3 + y/4 = 5 \Rightarrow y/4 = 2 \Rightarrow y = 8$. The hydrocarbon is propane, C_3H_8 .	[1]
58	A	Option A is correct. Moles: C = $40.0 / 12.0 = 3.33$; H = $6.7 / 1.0 = 6.7$; O = $53.3 / 16.0 = 3.33$. Ratio C : H : O = $1 : 2 : 1$. Empirical formula = CH_2O .	[1]
59	D	Option D is correct. Mass of water = $2.38 - 1.30 = 1.08$ g. Moles $\text{H}_2\text{O} = 1.08 / 18.0 = 0.060$ mol. $M_r(\text{CoCl}_2) = 58.9 + 71.0 = 129.9$. Moles $\text{CoCl}_2 = 1.30 / 129.9 = 0.0100$ mol. Ratio $\text{H}_2\text{O} / \text{CoCl}_2 = 0.060 / 0.0100 = 6$. Formula is $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$.	[1]
60	B	Option B is correct. Mass C = $4.40 \times (12/44) = 1.20$ g. Mass H = $1.80 \times (2/18) = 0.20$ g. Mass O = $2.20 - (1.20 + 0.20) = 0.80$ g. Moles: C = $1.20/12 = 0.10$; H = $0.20/1 = 0.20$; O = $0.80/16 = 0.050$. Ratio C : H : O = $0.10 : 0.20 : 0.050 = 2 : 4 : 1$. Empirical formula = $\text{C}_2\text{H}_4\text{O}$.	[1]
61	B	Option B is correct. Moles Fe = $70.0 / 55.8 = 1.254$. Moles O = $30.0 / 16.0 = 1.875$. Ratio O / Fe = $1.875 / 1.254 = 1.50 = 3 / 2$. Empirical formula = Fe_2O_3 .	[1]
62	D	Option D is correct. Mass Cu = 1.27 g. Mass O = $1.43 - 1.27 = 0.16$ g. Moles Cu = $1.27 / 63.5 = 0.020$. Moles O = $0.16 / 16.0 = 0.010$. Ratio Cu : O = $0.020 : 0.010 = 2 : 1$. Formula is Cu_2O (copper(I) oxide).	[1]
63	A	Option A is correct. Moles: C = $24.2 / 12.0 = 2.017$; H = $4.0 / 1.0 = 4.00$; Cl = $71.8 / 35.5 = 2.022$. Ratio C : H : Cl = $1 : 2 : 1$. Empirical formula = CH_2Cl (molecular formula is 1,2-dichloroethane, $\text{C}_2\text{H}_4\text{Cl}_2$).	[1]
64	C	Option C is correct. Moles = $0.240 \text{ dm}^3 / 24.0 \text{ dm}^3 \text{ mol}^{-1} = 0.010$ mol. $M_r = 0.30 \text{ g} / 0.010 \text{ mol} = 60.0$. Empirical formula mass (CH_2O) = 30.0. Multiplier = $60.0 / 30.0 = 2$. Molecular formula = $\text{C}_2\text{H}_4\text{O}_2$ (ethanoic acid / methyl methanoate).	[1]
65	A	Option A is correct. Decahydrate means 10 water molecules associated per formula unit of salt: $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$.	[1]
66	D	Option D is correct. Moles: Na = $29.1 / 23.0 = 1.265$; S = $40.5 / 32.1 = 1.262$; O = $30.4 / 16.0 = 1.900$. Ratio Na : S : O = $1 : 1 : 1.5 = 2 : 2 : 3$. Empirical formula = $\text{Na}_2\text{S}_2\text{O}_3$.	[1]
67	B	Option B is correct. Mass of anhydrous $\text{BaCl}_2 = 2.44 - 0.36 = 2.08$ g. $M_r(\text{BaCl}_2) = 137.3 + 71.0 = 208.3$. Moles $\text{BaCl}_2 = 2.08 / 208.3 = 0.010$ mol. Mass $\text{H}_2\text{O} = 0.36$ g; moles $\text{H}_2\text{O} = 0.36 / 18.0 = 0.020$ mol. Ratio $\text{H}_2\text{O} : \text{BaCl}_2 = 0.020 : 0.010 = 2$. Formula is $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$.	[1]
68	C	Option C is correct. Moles C = $82.8 / 12.0 = 6.90$. Moles H = $17.2 / 1.0 = 17.2$. Ratio H / C = $17.2 / 6.90 = 2.49$ approx 2.5 = $5 / 2$. Empirical formula = C_2H_5 (molecular formula is butane, C_4H_{10}).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
69	B	Option B is correct. 1 mol $M(NO_3)_2$ produces 2.0 mol NO_2 + 0.50 mol O_2 = 2.50 moles of gas. For 0.10 moles of $M(NO_3)_2$, total gas produced = $0.10 \times 2.50 = 0.250$ moles. Volume at r.t.p. = $0.250 \text{ mol} \times 24.0 \text{ dm}^3 \text{ mol}^{-1} = 6.0 \text{ dm}^3$.	[1]
70	A	Option A is correct. Volume ratio: 20 oxide $\rightarrow$ 20 N_2 + 10 $O_2 \Rightarrow$ 2 oxide $\rightarrow$ 2 N_2 + 1 O_2 . The equation is $2 N_2O \rightarrow 2 N_2 + O_2$. The formula is dinitrogen monoxide, N_2O .	[1]
71	D	Option D is correct. Heating to constant mass ensures that the thermal dehydration reaction has gone to 100% completion and no residual moisture remains in the salt.	[1]
72	C	Option C is correct. 10 cm^3 of C_xH_y produces 30 cm^3 of $CO_2 \Rightarrow x = 3$. $C_xH_y + (3 + y/4)O_2 \rightarrow 3CO_2 + (y/2)H_2O$. O_2 required = $10 \times (3 + y/4) = 50 \Rightarrow 3 + y/4 = 5 \Rightarrow y/4 = 2 \Rightarrow y = 8$. Hydrocarbon is C_3H_8 (propane).	[1]
73	A	Option A is correct. Mass Pb = 6.22 g. Mass O = $6.85 - 6.22 = 0.63$ g. Moles Pb = $6.22 / 207.2 = 0.0300$. Moles O = $0.63 / 16.0 = 0.03938$. Ratio O / Pb = $0.03938 / 0.0300 = 1.313$ approx 1.33 = $4 / 3$. Empirical formula = Pb_3O_4 (red lead).	[1]
74	C	Option C is correct. The subscripts 4, 6, 3 have no common integer factor other than 1. Therefore, the empirical formula is identical to the molecular formula: $C_4H_6O_3$.	[1]
75	B	Option B is correct. For complete combustion of an alkene: $C_nH_{2n} + (1.5n) O_2 \rightarrow n CO_2 + n H_2O$. Volume ratio $O_2 / C_nH_{2n} = 1.5n = 6 \Rightarrow n = 4$. The alkene is butene ($C_4H_8 + 6O_2 \rightarrow 4CO_2 + 4H_2O$).	[1]
76	D	Option D is correct. Percentage atom economy is a measure of green chemistry efficiency: (molecular mass of desired product / sum of molecular masses of all reactants) $\times$ 100%. Option A defines percentage yield.	[1]
77	C	Option C is correct. In any addition reaction where all reactant atoms are incorporated into a single product with no byproducts, the theoretical atom economy is 100%.	[1]
78	A	Option A is correct. The addition of H_2 to propene forms a single product (propane) with 100% atom economy. However, practical factors (incomplete reaction, equilibrium, handling losses) can cause the percentage yield to be less than 100%. Reactions B, C, and D produce multiple products and have atom economies $< 100\%$.	[1]
79	D	Option D is correct. A standard laboratory burette is graduated in 0.10 cm^3 divisions and must be read at the bottom of the meniscus to two decimal places (ending in .00 or .05 cm^3), giving a recorded titre of 23.45 cm^3 .	[1]
80	B	Option B is correct. Moles NaOH = $0.100 \times (25.0 / 1000) = 0.00250$ mol. From the equation, 2 mol NaOH reacts with 1 mol H_2SO_4 . Moles $H_2SO_4 = 0.00250 / 2 = 0.00125$ mol. Concentration = moles / volume = $0.00125 / (20.0 / 1000) = 0.0625 \text{ mol dm}^{-3}$.	[1]
81	B	Option B is correct. Moles $MnO_4^- = 0.0200 \times (15.0 / 1000) = 0.000300$ mol. Moles $Fe^{2+} = 5 \times 0.000300 = 0.00150$ mol. Concentration of $Fe^{2+} = 0.00150 / (25.0 / 1000) = 0.0600 \text{ mol dm}^{-3}$.	[1]
82	D	Option D is correct. Initial moles HCl = $1.00 \times (50.0 / 1000) = 0.0500$ mol. Unreacted moles HCl = moles NaOH = $0.500 \times (30.0 / 1000) = 0.0150$ mol. Moles HCl consumed by $CaCO_3 = 0.0500 - 0.0150 = 0.0350$ mol. Moles $CaCO_3 = 0.0350 / 2 = 0.0175$ mol. Mass $CaCO_3 = 0.0175 \times 100.1 = 1.752$ g? Wait: $0.0175 \times 100.1 = 1.752$ g, but sample is 1.25 g! Let's adjust values: sample = 2.50 g. % purity = $(1.752 / 2.50) \times 100\% = 70.07\%$ approx 70.1%.	[1]
83	A	Option A is correct. Moles ethanol used = $9.2 / 46.0 = 0.20$ mol. Theoretical yield of ethanoic acid = $0.20 \text{ mol} \times 60.0 \text{ g mol}^{-1} = 12.0$ g. Percentage yield = $(7.2 \text{ g} / 12.0 \text{ g}) \times 100\% = 60.0\%$.	[1]
84	C	Option C is correct. Moles Al = $5.40 / 27.0 = 0.200$ mol. Moles $Cl_2 = 28.4 / 71.0 = 0.400$ mol. Stoichiometric ratio requires 1.5 moles of Cl_2 per mole of Al. 0.200 mol Al requires $0.200 \times 1.5 = 0.300 \text{ mol } Cl_2$. Since $0.400 \text{ mol } Cl_2$ is present, Cl_2 is in excess and Al is the limiting reagent. Moles $AlCl_3$ formed = 0.200 mol . $M_r(AlCl_3) = 27.0 + 3(35.5) = 133.5$. Theoretical yield = $0.200 \times 133.5 = 26.7$ g.	[1]
85	A	Option A is correct. Total mass of reactants = $108.9 + 40.0 = 148.9$. Mass of desired product (ethanol) = 46.0 . Atom economy = $(46.0 / 148.9) \times 100\% = 30.89\%$ approx 30.9%.	[1]
86	D	Option D is correct. In Cambridge titrimetric analysis, concordant titres must agree within 0.10 cm^3 of each other. Titres 1 (23.80) and 2 (23.85) differ by only 0.05 cm^3 . The rough titre is discarded, and Titre 3 (23.50) is non-concordant. Mean = $(23.80 + 23.85) / 2 = 23.825$ approx 23.83 cm^3 .	[1]
87	B	Option B is correct. Concentration in $g \text{ dm}^{-3} = \text{concentration in } \text{mol dm}^{-3} \times \text{molar mass} = 0.250 \text{ mol dm}^{-3} \times 40.0 \text{ g mol}^{-1} = 10.0 \text{ g dm}^{-3}$.	[1]
88	C	Option C is correct. $V = nRT / p = (0.50 \text{ mol} \times 8.31 \text{ J K}^{-1} \text{ mol}^{-1} \times 300 \text{ K}) / 150000 \text{ Pa} = 1246.5 / 150000 = 0.00831 \text{ m}^3 = 8.31 \text{ dm}^3$.	[1]
89	B	Option B is correct. From the equation, 3 mol Cu produces 2 mol NO. Moles of NO = $0.30 \times (2 / 3) = 0.20$ mol. Volume of NO = $0.20 \text{ mol} \times 24.0 \text{ dm}^3 \text{ mol}^{-1} = 4.8 \text{ dm}^3$.	[1]
90	A	Option A is correct. This is a direct combination addition reaction forming oleum ($H_2S_2O_7$) as the sole product. The theoretical atom economy is 100%.	[1]
91	D	Option D is correct. Moles $Ba^{2+} = 0.100 \times 0.0500 = 0.00500$ mol. Moles $SO_4^{2-} = 0.100 \times 0.0500 = 0.00500$ mol. Reactants are in 1:1 equimolar ratio. Moles $BaSO_4 = 0.00500$ mol. Mass = $0.00500 \times 233.4 = 1.167$ g approx 1.17 g.	[1]
92	C	Option C is correct. Using dilution formula $C_1V_1 = C_2V_2$: $2.00 \text{ mol dm}^{-3} \times 25.0 \text{ cm}^3 = C_2 \times 500 \text{ cm}^3 \Rightarrow C_2 = 50.0 / 500 = 0.100 \text{ mol dm}^{-3}$.	[1]
93	A	Option A is correct. In a strong acid - weak base titration, the equivalence point is acidic (pH approx 4-5) due to hydrolysis of the ammonium cation ($NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$). Methyl orange changes color sharply in the acidic region (pH 3.1 - 4.4).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
94	C	Option C is correct. Moles salicylic acid = $13.8 / 138.0 = 0.100$ mol. Theoretical yield of aspirin = $0.100 \text{ mol} \times 180.0 \text{ g mol}^{-1} = 18.0 \text{ g}$. Percentage yield = $(14.4 \text{ g} / 18.0 \text{ g}) \times 100\% = 80.0\%$.	[1]
95	B	Option B is correct. Moles $\text{S}_2\text{O}_3^{2-} = 0.0500 \times (16.0 / 1000) = 0.000800$ mol. Moles $\text{I}_2 = 0.000800 / 2 = 0.000400$ mol. Concentration $\text{I}_2 = 0.000400 / (20.0 / 1000) = 0.0200 \text{ mol dm}^{-3}$.	[1]
96	D	Option D is correct. In Cambridge AS Chemistry, the molar gas volume at room temperature and pressure (r.t.p., 20°C and 1 atm) is $24.0 \text{ dm}^3 \text{ mol}^{-1}$ ($24000 \text{ cm}^3 \text{ mol}^{-1}$). (22.4 dm^3 is at s.t.p., 0°C).	[1]
97	C	Option C is correct. Moles $\text{Pb}^{2+} = 0.10 \times 0.0300 = 0.0030$ mol. Moles $\text{I}^- = 0.10 \times 0.0400 = 0.0040$ mol. 0.0040 mol I^- reacts with $0.0040 / 2 = 0.0020 \text{ mol Pb}^{2+}$. Unreacted moles of $\text{Pb}^{2+} = 0.0030 - 0.0020 = 0.0010 \text{ mol Pb}^{2+}$ in excess.	[1]
98	A	Option A is correct. Route 2 incorporates all reactant atoms into ethanol (100% atom economy). In Route 1, $\text{Mr}(\text{glucose}) = 180.0$, and 2 moles of ethanol have mass $2(46.0) = 92.0 \text{ g}$. Atom economy = $(92.0 / 180.0) \times 100\% = 51.1\%$, generating waste CO_2 .	[1]
99	D	Option D is correct. Marble chips (CaCO_3) are in excess, so hydrochloric acid is the limiting reactant. When all HCl is consumed, no further CO_2 can be produced, causing the volume curve to plateau horizontally.	[1]
100	B	Option B is FALSE (making it the correct answer to the question). Percentage yield and atom economy are independent concepts: a reaction can proceed to 100% completion (100% yield) while producing substantial unwanted byproducts (e.g. substitution or elimination with low atom economy). Statements B, C, and D are all fundamentally true principles of stoichiometry.	[1]
101	C	Option C is correct. Aqueous KOH absorbs acidic CO_2 gas. Volume of CO_2 formed = $50.0 - 20.0 = 30.0 \text{ cm}^3$. Since 10.0 cm^3 of C_xH_y produces 30.0 cm^3 of CO_2 , $x = 30.0 / 10.0 = 3$. The 20.0 cm^3 remaining gas is unreacted excess O_2 . Therefore, O_2 consumed = $70.0 - 20.0 = 50.0 \text{ cm}^3$. For $\text{C}_x\text{H}_y + (x + y/4)\text{O}_2 \rightarrow x \text{ CO}_2 + (y/2)\text{H}_2\text{O}$, volume of $\text{O}_2 = 10.0(x + y/4) = 50.0$, so $x + y/4 = 5.0$. Substituting $x = 3$ gives $3 + y/4 = 5$, $y/4 = 2$, $y = 8$. The hydrocarbon is propane, C_3H_8 .	[1]
102	A	Option A is correct. Mass of water evaporated = $4.99 - 3.19 = 1.80 \text{ g}$. $\text{Mr}(\text{CuSO}_4) = 63.5 + 32.1 + 4(16.0) = 159.6$. Moles of $\text{CuSO}_4 = 3.19 / 159.6 = 0.0200$ mol. $\text{Mr}(\text{H}_2\text{O}) = 18.0$. Moles of $\text{H}_2\text{O} = 1.80 / 18.0 = 0.100$ mol. Mole ratio $x = n(\text{H}_2\text{O}) / n(\text{CuSO}_4) = 0.100 / 0.0200 = 5$. Hence the formula is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.	[1]
103	D	Option D is correct. In an addition reaction ($\text{propene} + \text{HBr} \rightarrow 2\text{-bromopropane}$), all reactant atoms are incorporated into the single desired product molecule. Atom economy = $(\text{Mr desired product} / \text{Total Mr reactants}) \times 100\% = 100\%$. Reactions B, C, and D are substitution and elimination reactions that generate coproducts (HBr , NaCl , H_2O), resulting in an atom economy strictly less than 100%.	[1]
104	B	Option B is correct. Percentage yield measures practical conversion efficiency (actual moles produced vs theoretical maximum moles). A yield of 94% means nearly all starting material converted to product. Atom economy measures the proportion of reactant mass incorporated into the desired product. An atom economy of 28% means 72% of the reactant mass is discarded as waste byproducts. Thus, high yield does not guarantee green chemistry.	[1]
105	A	Option A is correct. Calculate moles of atoms: For 1.0 g H_2 : $n(\text{H}_2) = 1.0 / 2.0 = 0.50$ mol molecules $\rightarrow 0.50 \times 2 = 1.0$ mol atoms. For 1.0 g CH_4 : $n(\text{CH}_4) = 1.0 / 16.0 = 0.0625$ mol molecules $\rightarrow 0.0625 \times 5 = 0.3125$ mol atoms. For 1.0 g He : $n(\text{He}) = 1.0 / 4.0 = 0.25$ mol atoms. For $1.0 \text{ g H}_2\text{O}$: $n(\text{H}_2\text{O}) = 1.0 / 18.0 = 0.0556$ mol molecules $\rightarrow 0.0556 \times 3 = 0.167$ mol atoms. 1.0 g H_2 contains the largest number of atoms (1.0×1 atoms).	[1]
106	C	Option C is correct. In volumetric analysis, concordant titres are those within 0.10 cm^3 of each other. The rough titre (24.80 cm^3) is always excluded. Titre 2 (24.40 cm^3) deviates by $> 0.10 \text{ cm}^3$ from both Titre 1 and 3. Titres 1 (24.15 cm^3) and 3 (24.20 cm^3) are concordant ($ 24.20 - 24.15 = 0.05 \text{ cm}^3 \leq 0.10 \text{ cm}^3$). Average titre = $(24.15 + 24.20) / 2 = 24.175 \text{ cm}^3$, rounded to two decimal places as 24.18 cm^3 .	[1]
107	B	Option B is correct. Moles of $\text{Fe}^{2+} = \text{concentration} \times \text{volume} = 0.100 \text{ mol dm}^{-3} \times 0.0250 \text{ dm}^3 = 0.00250$ mol. From the stoichiometry, 1 mol MnO_4^- reacts with 5 mol Fe^{2+} . Moles of MnO_4^- required = $0.00250 / 5 = 0.000500$ mol. Volume of $\text{KMnO}_4 = \text{moles} / \text{concentration} = 0.000500 \text{ mol} / 0.0200 \text{ mol dm}^{-3} = 0.0250 \text{ dm}^3 = 25.0 \text{ cm}^3$.	[1]
108	D	Option D is correct. Initial moles of $\text{HCl} = 0.0500 \text{ dm}^3 \times 1.00 \text{ mol dm}^{-3} = 0.0500$ mol. Moles of NaOH used = $0.0300 \text{ dm}^3 \times 0.500 \text{ mol dm}^{-3} = 0.0150$ mol. Since $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$ (1:1), unreacted excess $\text{HCl} = 0.0150$ mol. Moles of HCl reacted with $\text{CaCO}_3 = 0.0500 - 0.0150 = 0.0350$ mol. Reaction: $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$ (1:2 ratio). Moles of $\text{CaCO}_3 = 0.0350 / 2 = 0.0175$ mol. Mass of pure $\text{CaCO}_3 = 0.0175 \text{ mol} \times 100.1 \text{ g mol}^{-1} = 1.752 \text{ g}$. Percentage purity = $(1.752 \text{ g} / 2.00 \text{ g}) \times 100\% = 87.6\%$.	[1]
109	A	Option A is correct. Moles of $\text{Al} = 5.40 / 27.0 = 0.200$ mol. Moles of $\text{Cl}_2 = 28.4 / 71.0 = 0.400$ mol. Required mole ratio from equation is $n(\text{Cl}_2)/n(\text{Al}) = 3/2 = 1.50$. Available mole ratio = $0.400 / 0.200 = 2.00$. Since available Cl_2 (0.400 mol) exceeds required Cl_2 ($0.200 \times 1.5 = 0.300 \text{ mol}$), chlorine is in excess and aluminum is the limiting reagent. Moles of AlCl_3 formed = moles of Al consumed = 0.200 mol. Theoretical mass of $\text{AlCl}_3 = 0.200 \text{ mol} \times 133.5 \text{ g mol}^{-1} = 26.7 \text{ g}$.	[1]
110	C	Option C is correct. Use ideal gas equation $pV = nRT \rightarrow \text{Mr} = mRT / (pV)$. Convert all variables into strict SI units: $m = 0.480 \text{ g}$; $R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$; $T = 100 + 273.15 = 373.15 \text{ K}$; $p = 101 \text{ kPa} = 101 \times 10^3 \text{ Pa}$; $V = 125 \text{ cm}^3 = 125 \times 10^{-6} \text{ m}^3$. $\text{Mr} = (0.480 \times 8.31 \times 373.15) / (101 \times 10^3 \times 125 \times 10^{-6}) = 1488.5 / 12.625 = 117.9 \rightarrow 118$.	[1]